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PAPER_1007: Deconfinement Phase Diagram with SCm Boundary

Abstract

We construct the T vs μ_B phase diagram for the hadron-QGP transition with an SCm phonon boundary. The phase transition at T_c = 1.5×10^{12} K is modified by the running coupling α_s(T) and the $S_{26}^{(3)}$ phonon response.

1. Phase Classification

Temperature	Phase
T < 0.9 T_c	Hadron (confined)
0.9 T_c $\leq$ T $\leq$ 1.1 T_c	Mixed (crossover)
T > 1.1 T_c	QGP (deconfined)

2. SCm Phonon Boundary

The on-resonance phonon factor $\Phi(\Gamma_0) = S_{26}^{(3)} = 0.095$ modulates the crossover width.

3. Implementation

File: `scm_qgp_dynamics.py`, class `DeconfinementPhaseDiagramCalc`. CP4 class #591.